

MDAP · UNIVERSITY OF MELBOURNE

Mapping Melbourne, Searching Emotions

Two projects. Maps for Melbourne's suburbs, and search for a research database

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Two projects, two very different audiences

Both running at the same time. One for ABP, one for FEEL lab



Suburban Futures

Who

Faculty of Architecture, Building and Planning

What

Interactive maps of Melbourne's suburbs

Mentors

Amanda Belton, Emily Fitzgerald



EMOTE

Who

Psychology researchers (FEEL Lab)

What

Search a research database by meaning

Mentor

Daniel



imgflip.com

JANE-CLARK.TUMBLR

Where is infrastructure thin today?

Growth-corridor suburbs are adding residents far faster than local facilities can keep pace

+375K

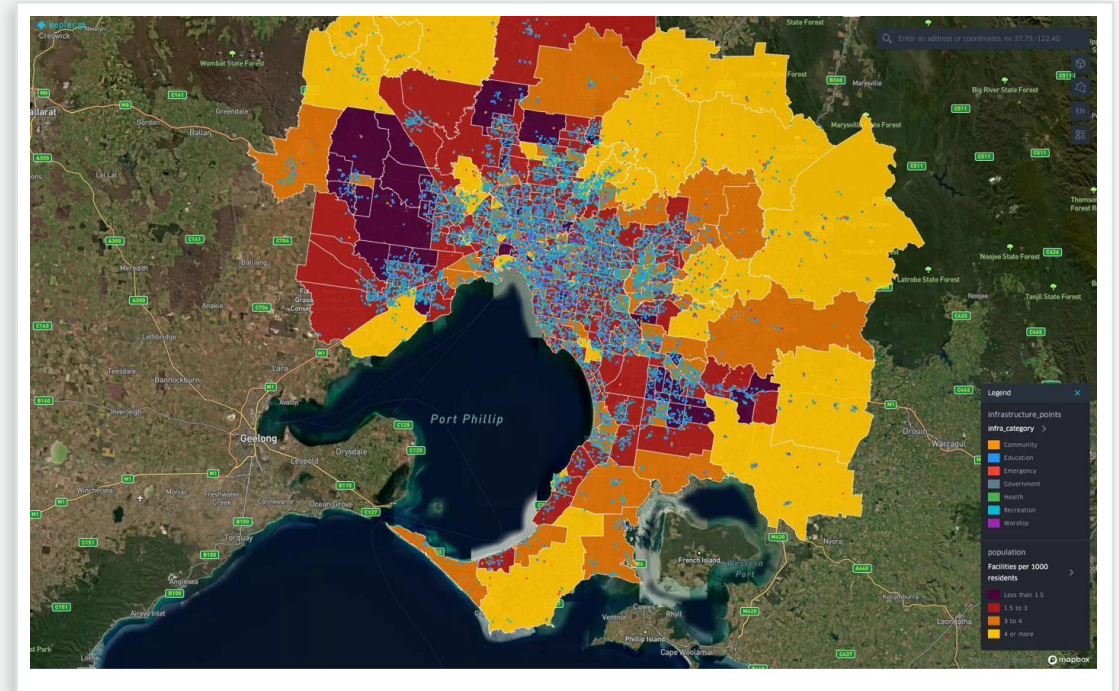
people added across Greater Melbourne, 2021 to 2024

25 SA2s

grew >20% yet sit below the metro median for facilities

0.39

facilities per 1,000 in Fraser Rise-Plumpton, which more than doubled in population (+102%)



Current Infrastructure Poverty. Facilities per 1,000 residents, 361 SA2s

Did it catch up? Partly, unevenly.

Melbourne added 10,494 facilities over 14 years. But in the fastest growing suburbs, people arrived so quickly the gap never really closed

+1.24M

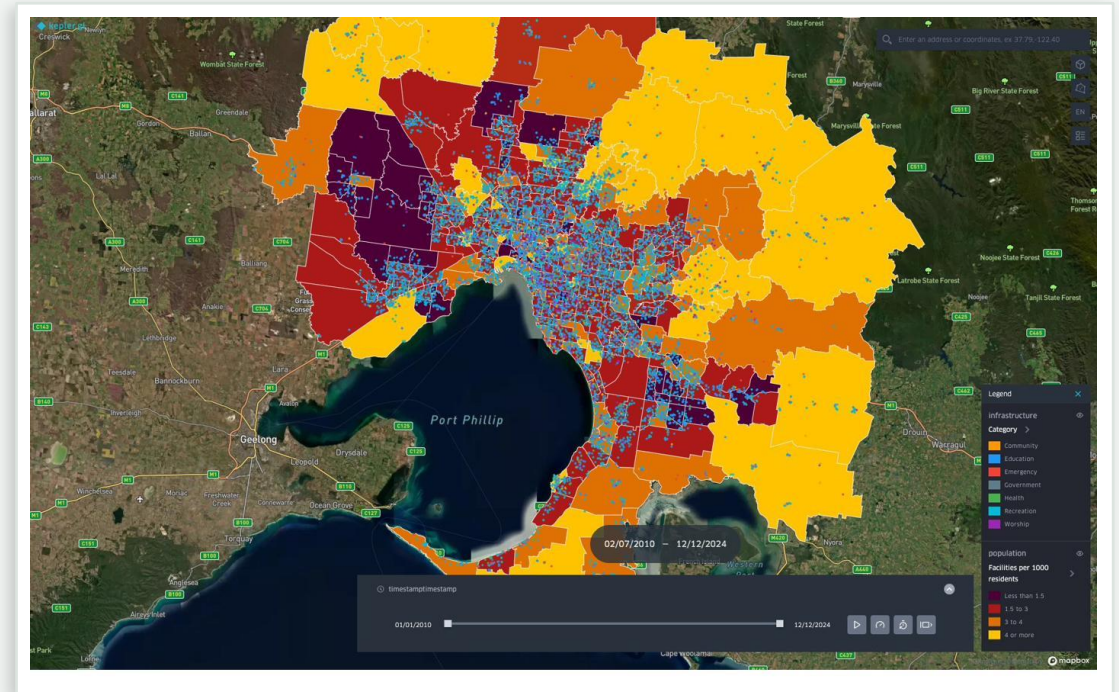
residents added city-wide, 2010 to 2024

+10,494

facilities added over the same window

1 per ~119

new facility for every 119 new residents over 14 years. Catch-up was real, but never fast enough.



Infrastructure Poverty Timeline. Facility growth vs population, 2010 to 2024

Which industry is driving the corridor surge?

Which industries are dominating which Melbourne suburbs in self employed businesses

The driver: Transport, Postal & Warehousing. Mostly gig and delivery work.

81%

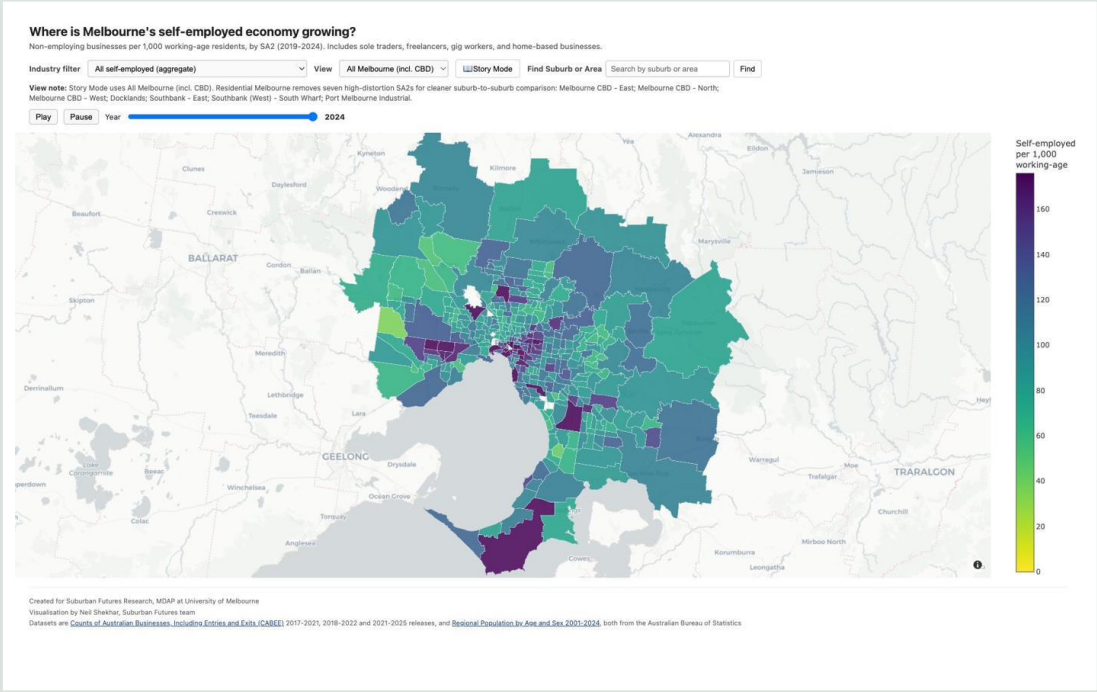
of metro Transport/delivery self-employed growth is in corridor LGAs

103 /1k

Tarneit-North leads on delivery businesses per 1,000 working age

0 overlap

between the top transport SA2s and the top construction SA2s. Two different corridor economies.



Self-Employed Business Density, 2019 to 2024 [▶ click image for live demo](#)

Who are these workers?

Map 3 could not say who the workers actually were. So I built Map 4 from Census data to answer that

69%

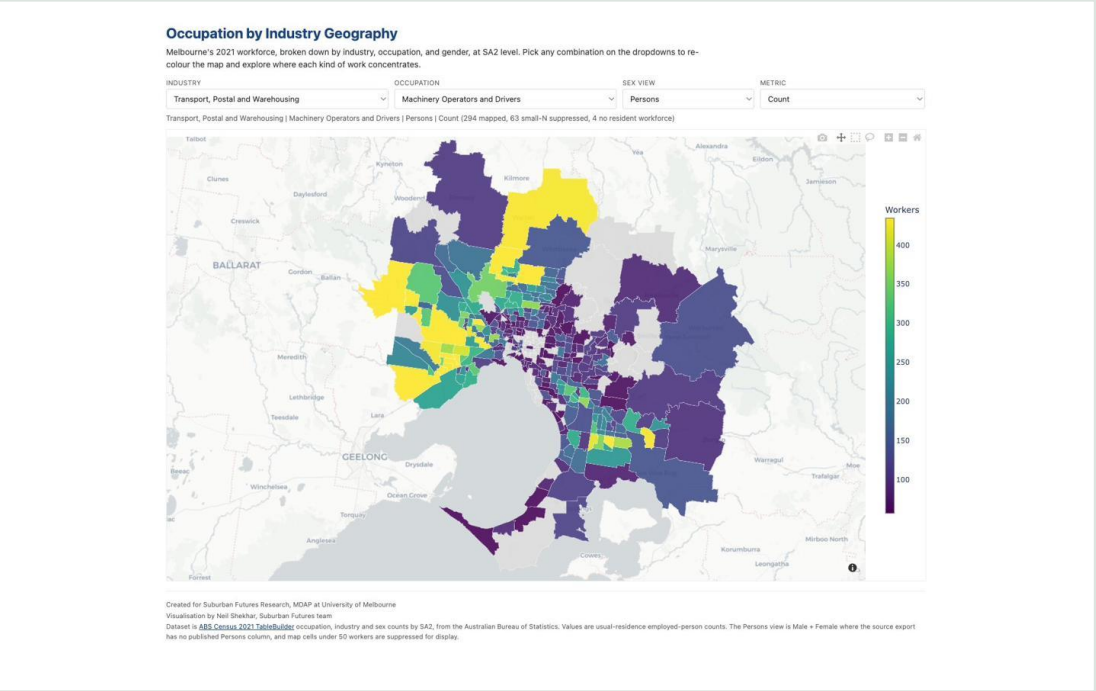
of Tarneit-North's transport workers are Machinery Operators & Drivers (379 of 552). The proof behind the delivery story.

77%

of Health Care & Social Assistance workers are women

13%

female share among Machinery Operators & Drivers, a sharp gender split

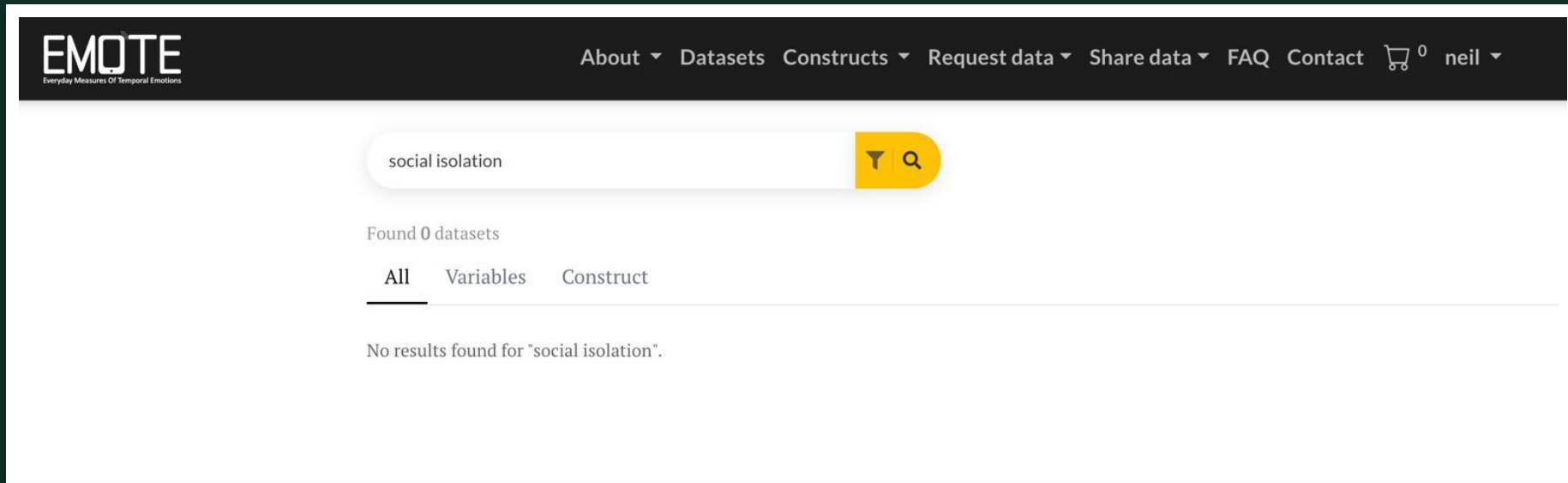


Occupation by Industry Geography, 2021 Census [▶ click image for live demo](#)

EMOTE · THE PROBLEM

Keyword search fails the moment the words don't match

A researcher searches "social isolation" and nothing matches



0 results

even though the data is right there. Keyword search just can't reach it.

Same query. Now it finds meaning, not just words

I built a semantic + hybrid search layer alongside the existing keyword search. "Social isolation" now returns the right variables instantly



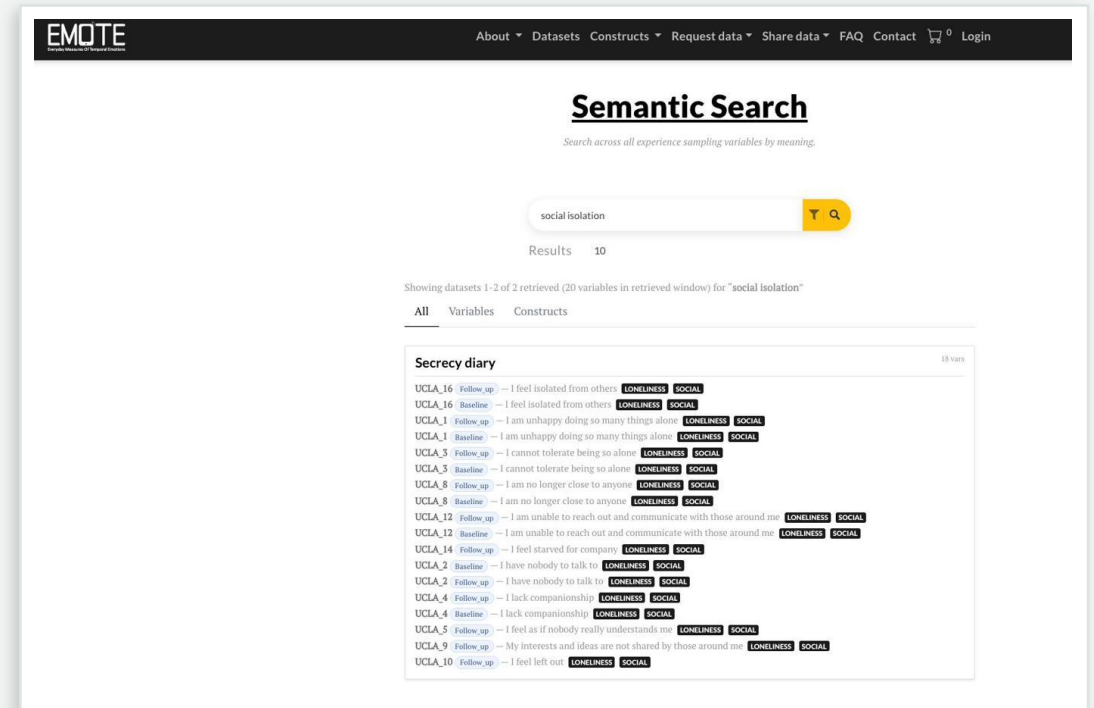
Three ways to search By meaning, by keyword, or both blended together



How meaning search works Text turned into number vectors, matched by closeness



Results in three tabs All, Variables, and Constructs, like EMOTE today



Hybrid search: "social isolation" now finds the UCLA loneliness items

All it needed was richer context, not a fancier model

On its own, a variable name is too thin for the search to place well. So I wrapped each one in the context around it: the study, the question, who took part. That is what made the ranking click.

Just the variable on its own

"feeling lonely at university"

ranked generic loneliness items above the ones that actually fit a student setting.



Variable plus its study context

"feeling lonely at university"

moved the first-year student study to the top, where it belongs.

From a paid API to a model we can run ourselves

I got it working fast with OpenAI's API, then tested a model we could run in-house. It held up, and it takes the running cost to zero.



Started with OpenAI

text-embedding-3-large

Quick to set up and a solid baseline. But it bills on every query, and that adds up over time.



Chose local Qwen

Qwen3-Embedding-4B

Held its own in my tests, with no per-query cost once it is running. Deployment is the next step.

HOW THE WORK ACTUALLY WENT

Research work is mostly trying things until one works.

The clean demos hide a lot of dead ends. That is where I learned the most.



1

Brainstormed widely

Three possible EMOTE projects, and several ways to frame the business map before one held up.



2

Failed, then diagnosed

Two models would not even load. Map visualization was breaking when loading data.



3

Then it worked

Built reusable tooling to test models early.

WHAT I TOOK FROM THE INTERNSHIP

What I am taking away

- Every tool here was new to me. I can pick up an unfamiliar stack and still ship something that works
- I learned to let the data set the story
- Working across two very different teams

Thank you

Neil Shekhar